On the Representation of a Large Even Integer as the Sum of a Prime and the Product of At Most Two Primes

Chen Jingrun*†

Abstract

The purpose of this paper is to prove, by the sieve method, that every sufficiently large even integer is the sum of a prime and a number which is either a prime or the product of at most two primes.

An analogous result is also obtained for the twin prime problem.

1 Introduction

Let the proposition "every sufficiently large even integer can be written as the sum of a prime and a number that is the product of at most a primes" be abbreviated as $(1, a)$.

Many mathematicians have improved the sieve method and certain results on the distribution of primes, and used these to improve $(1, a)$. We now briefly describe the historical development of $(1, a)$ as follows:

- $(1, c)$ — Renyi[1],
- $(1, 5)$ — Pan Chengdong[2], Barban[3],
- $(1, 4)$ — Wang Yuan[4], Pan Chengdong[5], Barban[6],
- $(1, 3)$ — Buchstab[7], Vinogradov[8], Bombieri[9],

In reference [10] we gave an outline of the proof of $(1, 2)$.

Let $P_x(1, 2)$ denote the number of primes p satisfying the following condition:

$$x - p = p_1 \quad \text{or} \quad x - p = p_2 p_3,$$

where p_1, p_2, p_3 are all primes.

Let x denote a sufficiently large even integer. Let

$$C_x = \prod_{\substack{p \mid x \\ p > 2}} \frac{p-1}{p-2} \prod_{p>2} \left(1 - \frac{1}{(p-1)^2}\right).$$

*(Institute of Mathematics, Academia Sinica)
†Received March 13, 1973.

For any given even integer h and sufficiently large x, let $x_h(1,2)$ denote the number of primes p satisfying the following condition:

$$p \leqslant x, \quad p+h=p_1 \quad \text{or} \quad p+h=p_2p_3,$$

where p_1, p_2, p_3 are all primes.

The purpose of this paper is to prove and improve all the results mentioned by the author in reference [10]; these are stated in detail as follows.

Theorem 1. $(1,2)$ *holds, and* $P_x(1,2) \geqslant \dfrac{0.67xC_x}{(\log x)^2}$.

Theorem 2. *For any even integer h, there exist infinitely many primes p such that $p+h$ has at most 2 prime factors, and*

$$x_h(1,2) \geqslant \frac{0.67xC_x}{(\log x)^2}.$$

In proving Theorem 1, the main tools are Lemmas 8 and 9 of this paper. In proving Lemma 8 we use a relatively straightforward numerical computation; in proving Lemma 9 we use Bombieri's theorem[9] together with a result of Richert[11].

2 Several Lemmas

Lemma 1. *Suppose $y \geqslant 0$, and let $[\log x]$ denote the integer part of $\log x$, $x>1$,*

$$\Phi(y) = \frac{1}{2\pi i}\int_{2-i\infty}^{2+i\infty} \frac{y^\omega \, d\omega}{\omega\left(1+\dfrac{\omega}{(\log x)^{1.1}}\right)^{[\log x]+1}}.$$

It is clear that $\Phi(y)=0$ when $0 \leqslant y \leqslant 1$. For all $y \geqslant 0$, $\Phi(y)$ is a non-decreasing function. When $\log x \geqslant 10^4$ and $y \geqslant e^{2(\log x)^{-0.1}}$, we have

$$1 - x^{-0.1} \leqslant \Phi(y) \leqslant 1.$$

Proof. We first prove that

$$\frac{\partial^r}{\partial \omega^r}\left(\frac{y^\omega}{\omega}\right) = \left(\frac{y^\omega}{\omega}\right)\left\{(\log y)^r + \sum_{i=1}^{r} \frac{(-1)^i r \cdots (r-i+1)(\log y)^{r-i}}{\omega^i}\right\} \tag{1}$$

holds. Clearly, formula (1) holds for $r=1$ and $r=2$. Now suppose it holds for $r=$

$2, \cdots, S$, and we prove it for $S+1$. Since

$$\frac{\partial^{S+1}}{\partial \omega^{S+1}}\left(\frac{y^\omega}{\omega}\right) = \frac{\partial}{\partial \omega}\left\{ y^\omega \left(\frac{(\log y)^S}{\omega} + \sum_{i=1}^{S} \frac{(-1)^i S \cdots (S-i+1)(\log y)^{S-i}}{\omega^{i+1}} \right) \right\}$$

$$= y^\omega \left\{ \frac{(\log y)^{S+1}}{\omega} + \sum_{i=1}^{S} \frac{(-1)^i S \cdots (S-i+1)(\log y)^{S+1-i}}{\omega^{i+1}} - \frac{(\log y)^S}{\omega^2} \right.$$

$$\left. + \sum_{i=1}^{S} \frac{(-1)^{i+1} S \cdots (S-i+1)(i+1)(\log y)^{S-i}}{\omega^{i+2}} \right\}$$

$$= \left(\frac{y^\omega}{\omega}\right) \left\{ (\log y)^{S+1} - \frac{(S+1)(\log y)^S}{\omega} + \frac{(-1)^{S+1}(S+1)!}{\omega^{S+1}} \right.$$

$$\left. + \sum_{i=2}^{S} \left(\frac{(-1)^i S \cdots (S-i+1)(\log y)^{S+1-i}}{\omega^i} + \frac{(-1)^i S \cdots (S+2-i) i (\log y)^{S+1-i}}{\omega^i} \right) \right\}$$

$$= \left(\frac{y^\omega}{\omega}\right) \left\{ (\log y)^{S+1} + \sum_{i=1}^{S+1} \frac{(-1)^i (S+1) \cdots (S+1-i+1)(\log y)^{S+1-i}}{\omega^i} \right\}.$$

This proves (1).

Also, when $y \geqslant 1$, we have

$$\Phi(y) = 1 + \left\{ \frac{(\log x)^{1.1+1.1[\log x]}}{[\log x]!} \right\} \left\{ \frac{\partial^{[\log x]}}{\partial \omega^{[\log x]}} \left(\frac{y^\omega}{\omega} \right) \right\}_{\omega = -(\log x)^{1.1}}$$

$$= 1 - e^{-(\log x)^{1.1}(\log y)} \sum_{\nu=0}^{[\log x]} \frac{\{(\log x)^{1.1}(\log y)\}^\nu}{\nu!}$$

$$= \left\{ \frac{1}{[\log x]!} \right\} \int_0^{(\log x)^{1.1}(\log y)} e^{-\lambda} \lambda^{[\log x]} d\lambda.$$

Since $\Phi(y) = 0$ when $0 \leqslant y \leqslant 1$, it follows from the above that $\Phi(y)$ is a non-decreasing function for $y \geqslant 0$. Furthermore, when $y \geqslant e^{2(\log x)^{-0.1}}$, we have

$$0 < 1 - \Phi(y) = \left\{ \frac{1}{[\log x]!} \right\} \int_{(\log x)^{1.1}(\log y)}^\infty e^{-\lambda} \lambda^{[\log x]} d\lambda$$

$$\leqslant \left\{ \frac{1}{[\log x]!} \right\} \int_{2[\log x]}^\infty e^{-\lambda} \lambda^{[\log x]} d\lambda = \left\{ \frac{\{[\log x]\}^{1+[\log x]}}{[\log x]!} \right\}$$

$$\cdot \int_2^\infty e^{-\lambda[\log x]} \lambda^{[\log x]} d\lambda = \left\{ \frac{e^{-[\log x]} \{[\log x]\}^{1+[\log x]}}{[\log x]!} \right\}$$

$$\cdot \int_1^\infty e^{-\lambda[\log x]} (1+\lambda)^{[\log x]} d\lambda \leqslant x^{-0.1}.$$

Here we have used $\log x \geqslant 10^4$ and, for $\lambda \geqslant 1$, $e^{\log(1+\lambda)} \leqslant e^{\lambda \log 2}$. □

Lemma 2. *Let* $e(\alpha) = e^{2\pi i \alpha}$, $S(\alpha) = \sum_{n=M+1}^{M+N} a_n e(n\alpha)$, $Z = \sum_{n=M+1}^{M+N} |a_n|^2$, *where* a_n *are arbitrary real numbers. We write* $\sum_{\chi_q}^{*}$ *for a sum taken over all, and only, the primitive* *characters mod q. Then we have*

$$\sum_{q\leqslant X}\frac{q}{\varphi(q)}\sum_{\chi_q}{}^{*}\left|\sum_{n=M+1}^{M+N}a_n\chi_q(n)\right|^2\leqslant\left(X^2+\pi N\right)\sum_{n=M+1}^{M+N}|a_n|^2; \tag{2}$$

$$\sum_{D<q\leqslant Q}\frac{1}{\varphi(q)}\sum_{\chi_q}{}^{*}\left|\sum_{n=M+1}^{M+N}a_n\chi_q(n)\right|^2\ll\left(Q+\frac{N}{D}\right)\sum_{n=M+1}^{M+N}|a_n|^2. \tag{3}$$

Proof. Let F be a complex-valued differentiable function of period 1. Then

$$\left|F\left(\frac{a}{q}\right)\right|=\left|F(\alpha)-\int_{\frac{a}{q}}^{\alpha}dF(\beta)\right|\leqslant|F(\alpha)|+\int_{\frac{a}{q}}^{\alpha}|F'(\beta)||d\beta|,$$

Let $I(a,q)$ denote the interval of length $\frac{1}{Q^2}$ centered at $\frac{a}{q}$. It is clear that when $1\leqslant a<q$, $(a,q)=1$, $q\leqslant Q$, all the intervals $I(a,q)$ are pairwise disjoint, so

$$\begin{aligned}\sum_{q\leqslant Q}\sum_{\substack{a=1\\(a,q)=1}}^{q}\left|F\left(\frac{a}{q}\right)\right|&\leqslant\sum_{q\leqslant Q}\sum_{\substack{a=1\\(a,q)=1}}^{q}\left\{Q^2\int_{I(a,q)}|F(\alpha)|d\alpha+\frac{1}{2}\int_{I(a,q)}|F'(\beta)|d\beta\right\}\\&\leqslant Q^2\int_0^1|F(\alpha)|d\alpha+\frac{1}{2}\int_0^1|F'(\beta)|d\beta.\end{aligned}$$

Taking $F(\alpha)=\{S(\alpha)\}^2$, we obtain

$$\int_0^1|F(\alpha)|d\alpha=Z$$

and

$$\begin{aligned}\frac{1}{2}\int_0^1|F'(\beta)|d\beta&=\int_0^1|S(\alpha)||S'(\alpha)|d\alpha\\&\leqslant\left\{\left(\int_0^1|S(\alpha)|^2d\alpha\right)\left(\int_0^1|S'(\alpha)|^2d\alpha\right)\right\}^{\frac{1}{2}}=Z^{\frac{1}{2}}\left(\int_0^1|S'(\alpha)|^2d\alpha\right)^{\frac{1}{2}}.\end{aligned}$$

Hence

$$\begin{aligned}
\sum_{q\leqslant Q}\sum_{\substack{a=1\\(a,q)=1}}^{q}\left|S\left(\frac{a}{q}\right)\right|^2 &= \sum_{q\leqslant Q}\sum_{\substack{a=1\\(a,q)=1}}^{q}\left\{\left|S\left(\frac{a}{q}\right)\right| e\left(-\frac{a\left(M+\left[\frac{N}{2}\right]\right)}{q}\right)\right\}^2\\
&= \sum_{q\leqslant Q}\sum_{\substack{a=1\\(a,q)=1}}^{q}\left|\sum_{n=M+1}^{M+N} a_n e\left(\left\{n-\left(M+\left[\frac{N}{2}\right]\right)\right\}\frac{a}{q}\right)\right|^2\\
&= \sum_{q\leqslant Q}\sum_{\substack{a=1\\(a,q)=1}}^{q}\left|\sum_{-\left[\frac{N}{2}\right]+1\leqslant n\leqslant N-\left[\frac{N}{2}\right]} a_{n+M+\left[\frac{N}{2}\right]} e\left(\frac{na}{q}\right)\right|^2\\
&\leqslant ZQ^2 + Z^{\frac{1}{2}}\left\{\sum_{n=-\left[\frac{N}{2}\right]+1}^{N-\left[\frac{N}{2}\right]}\left((2\pi n)a_{n+M+\left[\frac{N}{2}\right]}\right)^2\right\}^{\frac{1}{2}}\\
&\leqslant ZQ^2 + \pi N Z^{\frac{1}{2}}\left(\sum_{n=-\left[\frac{N}{2}\right]+1}^{N-\left[\frac{N}{2}\right]}\left|a_{n+M+\left[\frac{N}{2}\right]}\right|^2\right)^{\frac{1}{2}}\\
&\leqslant (Q^2+\pi N)Z. \qquad (4)
\end{aligned}$$

Let χ^* denote a primitive character,

$$\tau(\chi_q^*) = \sum_{1\leqslant a<q}\chi_q^*(a)e\left(\frac{a}{q}\right), \qquad \tau(\overline{\chi_q^*})\chi_q^*(n) = \sum_{a=1}^{q}\overline{\chi_q^*}(a)e\left(\frac{na}{q}\right).$$

Since $|\tau(\overline{\chi_q^*})|^2 = q$, we obtain

$$\begin{aligned}
\left(\frac{1}{\varphi(q)}\right)\sum_{\chi_q}{}^{*}\left|\sum_{n=M+1}^{M+N} a_n\chi_q(n)\right|^2 &\leqslant \left(\frac{1}{q\varphi(q)}\right)\sum_{\chi_q}{}^{*}\left|\tau(\overline{\chi_q})\sum_{n=M+1}^{M+N} a_n\chi_q(n)\right|^2\\
&= \left(\frac{1}{q\varphi(q)}\right)\sum_{\chi_q}{}^{*}\left|\sum_{a=1}^{q}\overline{\chi_q}(a)\sum_{n=M+1}^{M+N} a_n e\left(\frac{na}{q}\right)\right|^2\\
&\leqslant \left(\frac{1}{q\varphi(q)}\right)\sum_{\chi_q}\left|\sum_{a=1}^{q}\overline{\chi_q}(a)\sum_{n=M+1}^{M+N} a_n e\left(\frac{na}{q}\right)\right|^2\\
&\leqslant \frac{1}{q}\sum_{\substack{a=1\\(a,q)=1}}^{q}\left|\sum_{n=M+1}^{M+N} a_n e\left(\frac{na}{q}\right)\right|^2.
\end{aligned}$$

From this and (4), we obtain (2). Let h be the positive integer such that $2^h D < Q \leqslant$

$2^{h+1}D$; then we have

$$\begin{aligned}\sum_{D<q\leqslant Q}\frac{1}{\varphi(q)}\sum_{\chi_q}{}^{*}\left|\sum_{n=M+1}^{M+N}a_n\chi_q(n)\right|^2 &\leqslant \sum_{i=0}^{h}\left(\sum_{2^iD<q\leqslant 2^{i+1}D}\frac{1}{\varphi(q)}\sum_{\chi_q}{}^{*}\left|\sum_{n=M+1}^{M+N}a_n\chi_q(n)\right|^2\right)\\ &\leqslant \sum_{i=0}^{h}\left(\frac{1}{2^iD}\right)\left(\sum_{2^iD<q\leqslant 2^{i+1}D}\frac{q}{\varphi(q)}\sum_{\chi_q}{}^{*}\left|\sum_{n=M+1}^{M+N}a_n\chi_q(n)\right|^2\right)\\ &\leqslant \sum_{i=0}^{h}\left(2^{i+2}D+\frac{\pi N}{2^iD}\right)\sum_{n=M+1}^{M+N}|a_n|^2\\ &\ll \left(Q+\frac{N}{D}\right)\sum_{n=M+1}^{M+N}|a_n|^2.\end{aligned}$$

This proves Lemma 2. □

Lemma 3. *When $S=\sigma+it$ and $\sigma\geqslant\frac{1}{2}$, we have*

$$\sum_{q\leqslant Q}\sum_{\chi_q}{}^{*}|L(S,\chi_q)|^4\ll Q^2|S|^2(\log Q)^4.$$

Proof. We have

$$\begin{aligned}L(S,\chi)=\sum_{n=1}^{\infty}\frac{\chi(n)}{n^S}&=\sum_{n=1}^{N}\frac{\chi(n)}{n^S}+\sum_{n=N+1}^{\infty}\frac{\sum_{i\leqslant n}\chi(i)-\sum_{i\leqslant n-1}\chi(i)}{n^S}\\ &=\sum_{n=1}^{N}\frac{\chi(n)}{n^S}+\sum_{n=N+1}^{\infty}\left(\sum_{i\leqslant n}\chi(i)\right)\left(\frac{1}{n^S}-\frac{1}{(n+1)^S}\right)-\frac{\sum_{i\leqslant N}\chi(i)}{(N+1)^S}\\ &=\sum_{n=1}^{N}\frac{\chi(n)}{n^S}+O\left(\frac{|S|q^{\frac12}\log q}{N^\sigma}\right).\end{aligned}$$

Hence, by Lemma 2 and $\sigma\geqslant\frac{1}{2}$, we have

$$\begin{aligned}\sum_{q\leqslant Q}\sum_{\chi_q}{}^{*}|L(S,\chi_q)|^4&\ll\sum_{q\leqslant Q}\sum_{\chi_q}{}^{*}\left(\left|\sum_{n=1}^{[Q|S|]}\frac{\chi_q(n)}{n^S}\right|^4+Q^{-2}|S|^2q^2(\log q)^4\right)\\ &\ll |S|^2Q^2(\log Q)^4+\left(Q^2+Q^2|S|^2\right)\sum_{n=1}^{[Q|S|]^2}\frac{d^2(n)}{n}\\ &\ll Q^2|S|^2(\log Q)^4.\end{aligned}$$

This proves the lemma. □

Lemma 4. *When k is odd and squarefree, and $m\neq 1$, we have*

$$\left|\sum_{\chi_k}{}^{*}\chi_k(m)\right|\leqslant|(m-1,k)|.$$

Proof. Let $k = p_1 \cdots p_l$, where $p_1 < \cdots < p_l$. Let g_i be a primitive root mod p_i, so that $m \equiv g_i^{\xi_i} \pmod{p_i}$, $0 \leqslant \xi_i \leqslant p_i - 2$, for $j = 1, \cdots, l$; then all primitive characters mod k can be written as

$$\chi_k^*(m) = e^{2\pi i\left(\frac{\nu_1\xi_1}{p_1-1}+\cdots+\frac{\nu_l\xi_l}{p_l-1}\right)},$$

where $1 \leqslant \nu_i \leqslant p_i - 2$, for $i = 1, \cdots, l$.

Let $Z(m,k) = \left|\sum_{\chi_k}^{*} \chi_k(m)\right|$; then

$$Z(m,k) = \prod_{j=1}^{l} Z(m,p_j) = \prod_{j=1}^{l}\left|\sum_{\nu_j=1}^{p_j-2} e^{2\pi i\frac{\nu_j\xi_j}{p_j-1}}\right| = \prod_{\substack{j=1\\ \xi_j=0}}^{l}(p_j-2) < \prod_{p_j|(m-1)} p_j = |(m-1,k)|.$$

This proves the lemma. □

Let x be an even integer, let $\lambda_1 = 1$; when $d > x^{\frac{1}{4}-\frac{\epsilon}{2}}$, let $\lambda_d = 0$; and when $1 < d \leqslant x^{\frac{1}{4}-\frac{\epsilon}{2}}$, let

$$\lambda_d = \frac{\mu(d)}{f(d)g(d)}\left\{\sum_{\substack{1\leqslant k\leqslant (x^{\frac{1}{2}-\epsilon})^{\frac{1}{2}}/d\\ (k,xd)=1}} \frac{\mu^2(k)}{f(k)}\right\}\left\{\sum_{\substack{1\leqslant k\leqslant (x^{\frac{1}{2}-\epsilon})^{\frac{1}{2}}\\ (k,x)=1}} \frac{\mu^2(k)}{f(k)}\right\}^{-1}.$$

where $g(k) = \dfrac{1}{\varphi(k)}$, $f(k) = \varphi(k)\prod_{p|k}\dfrac{p-2}{p-1}$. Furthermore, when d is odd and $\mu(d) \neq 0$, we have

$$\sum_{\substack{1\leqslant k\leqslant (x^{\frac{1}{2}-\epsilon})^{\frac{1}{2}}\\ (k,x)=1}} \frac{\mu^2(k)}{f(k)} = \sum_{t|d}\sum_{\substack{1\leqslant k\leqslant (x^{\frac{1}{2}-\epsilon})^{\frac{1}{2}}\\ (k,x)=1,\,(k,d)=t}} \frac{\mu^2(k)}{f(k)} = \sum_{t|d}\left\{\frac{1}{\prod_{p|t}(p-2)}\right\}\sum_{\substack{1\leqslant k\leqslant (x^{\frac{1}{2}-\epsilon})^{\frac{1}{2}}/t\\ (k,xd)=1}} \frac{\mu^2(k)}{f(k)}$$

$$\geqslant \left\{\prod_{p|d}\left(1+\frac{1}{p-2}\right)\right\}\left\{\sum_{\substack{1\leqslant k\leqslant (x^{\frac{1}{2}-\epsilon})^{\frac{1}{2}}/d\\ (k,xd)=1}} \frac{\mu^2(k)}{f(k)}\right\}.$$

Hence $|\lambda_d| \leqslant 1$ for all positive integers d. Let x be even, $\log x > 10^4$, and further let

$$Q = \prod_{2\leqslant p\leqslant x^{\frac{1}{4}}} p,$$

$$\Omega = \sum_{\substack{x^{\frac{1}{10}}<p_1\leqslant x^{\frac{1}{3}}<p_2\leqslant(\frac{x}{p_1})^{\frac{1}{2}}\\ p_3\leqslant\frac{x}{p_1p_2}\\ (x-p_1p_2p_3,Q)=1}} 1,$$

$$M = \sum_{x^{\frac{1}{10}}<p_1\leqslant x^{\frac{1}{3}}<p_2\leqslant(\frac{x}{p_1})^{\frac{1}{2}}}\left(\frac{1}{\log\dfrac{x}{p_1p_2}}\right)\left(\sum_{\substack{n\leqslant\frac{x}{p_1p_2}\\ (x-p_1p_2n,Q)=1}}\Lambda(n)\right),$$

then $\Omega \leqslant \dfrac{M}{1-\epsilon} + N$, where

$$N \ll \sum_{x^{\frac{1}{10}} < p_1 \leqslant x^{\frac{1}{3}} < p_2 \leqslant (\frac{x}{p_1})^{\frac{1}{2}}} \left(\frac{x}{p_1 p_2}\right)^{1-\epsilon} \ll x^{1-\epsilon} \int_{x^{\frac{1}{10}}}^{x^{\frac{1}{3}}} \frac{dS}{S^{1-\epsilon}} \int_{x^{\frac{1}{3}}}^{(\frac{x}{S})^{\frac{1}{2}}} \frac{dt}{t^{1-\epsilon}}$$
$$\ll x^{1-\frac{\epsilon}{2}} \int_{x^{\frac{1}{10}}}^{x^{\frac{1}{3}}} \frac{dS}{S^{1-\frac{\epsilon}{2}}} \ll x^{1-\frac{\epsilon}{3}}.$$

By Lemma 1, we have

$$\begin{aligned} M \leqslant & \sum_{x^{\frac{1}{10}} < p_1 \leqslant x^{\frac{1}{3}} < p_2 \leqslant (\frac{x}{p_1})^{\frac{1}{2}}} \left(\frac{1}{\log \dfrac{x}{p_1 p_2}}\right) \sum_{\substack{n \leqslant \frac{x}{p_1 p_2} \\ (x - p_1 p_2 n, Q) = 1}} \Lambda(n) \Phi\left(\frac{x}{p_1 p_2 n}\right) + O\left(\frac{x}{(\log x)^{2.01}}\right) \\ \leqslant & \sum_{x^{\frac{1}{10}} < p_1 \leqslant x^{\frac{1}{3}} < p_2 \leqslant (\frac{x}{p_1})^{\frac{1}{2}}} \left(\frac{1}{\log \dfrac{x}{p_1 p_2}}\right) \sum_{n \leqslant \frac{x}{p_1 p_2}} \Lambda(n) \Phi\left(\frac{x}{p_1 p_2 n}\right) \left(\sum_{\substack{d \mid (x - p_1 p_2 n, Q) \\ (d, x) = 1}} \lambda_d\right)^2 + O\left(\frac{x}{(\log x)^{2.01}}\right) \\ = & \sum_{\substack{(d_1, x) = 1 \\ d_1 \mid Q}} \sum_{\substack{(d_2, x) = 1 \\ d_2 \mid Q}} \lambda_{d_1} \lambda_{d_2} N_{\frac{d_1 d_2}{(d_1, d_2)}} + O\left(\frac{x}{(\log x)^{2.01}}\right). \end{aligned} \tag{5}$$

where

$$N_{\frac{d_1d_2}{(d_1,d_2)}} = \sum_{x^{\frac{1}{10}}<p_1\leqslant x^{\frac{1}{3}}<p_2\leqslant(\frac{x}{p_1})^{\frac{1}{2}}} \left(\frac{1}{\log\dfrac{x}{p_1p_2}}\right) \sum_{\substack{n\leqslant\frac{x}{p_1p_2}\\ x-p_1p_2n\equiv 0\left(\bmod \frac{d_1d_2}{(d_1,d_2)}\right)}} \Lambda(n)\Phi\left(\frac{x}{p_1p_2n}\right)$$

$$= \left\{\frac{1}{\varphi\left(\dfrac{d_1d_2}{(d_1,d_2)}\right)}\right\} \sum_{\substack{x^{\frac{1}{10}}<p_1\leqslant x^{\frac{1}{3}}<p_2\leqslant(\frac{x}{p_1})^{\frac{1}{2}}\\ n\leqslant\frac{x}{p_1p_2}\\ (p_1p_2n,d_1d_2)=1}} \left(\frac{1}{\log\dfrac{x}{p_1p_2}}\right)\Lambda(n)\Phi\left(\frac{x}{p_1p_2n}\right)$$

$$+ \sum_{\chi_{\frac{d_1d_2}{(d_1,d_2)}}\neq\chi_0} \overline{\chi_{\frac{d_1d_2}{(d_1,d_2)}}}(x) \sum_{\substack{x^{\frac{1}{10}}<p_1\leqslant x^{\frac{1}{3}}<p_2\leqslant(\frac{x}{p_1})^{\frac{1}{2}}\\ n\leqslant\frac{x}{p_1p_2}}} \left(\frac{\Lambda(n)}{\log\dfrac{x}{p_1p_2}}\right)\Phi\left(\frac{x}{p_1p_2n}\right)\chi_{\frac{d_1d_2}{(d_1,d_2)}}(p_1p_2n)$$

$$= \left\{\frac{1}{\varphi\left(\dfrac{d_1d_2}{(d_1,d_2)}\right)}\right\}\left\{\sum_{\substack{x^{\frac{1}{10}}<p_1\leqslant x^{\frac{1}{3}}<p_2\leqslant(\frac{x}{p_1})^{\frac{1}{2}}\\ n\leqslant\frac{x}{p_1p_2}\\ (p_1p_2n,d_1d_2)=1}} \left(\frac{1}{\log\dfrac{x}{p_1p_2}}\right)\Lambda(n)\Phi\left(\frac{x}{p_1p_2n}\right)\right\} - \left\{\frac{1}{2\pi i\,\varphi\left(\dfrac{d_1d_2}{(d_1,d_2)}\right)}\right\}$$

$$\cdot\left\{\int_{2-i\infty}^{2+i\infty}\left(1+\frac{\omega}{(\log x)^{1.1}}\right)^{-[\log x]-1}\left(\frac{x^\omega}{\omega}\right)\sum_{\chi_{\frac{d_1d_2}{(d_1,d_2)}}\neq\chi_0}\overline{\chi_{\frac{d_1d_2}{(d_1,d_2)}}}(x)\frac{L'}{L}(\omega,\chi_{\frac{d_1d_2}{(d_1,d_2)}})\right.$$

$$\left.\cdot\sum_{x^{\frac{1}{10}}<p_1\leqslant x^{\frac{1}{3}}<p_2\leqslant(\frac{x}{p_1})^{\frac{1}{2}}}\chi_{\frac{d_1d_2}{(d_1,d_2)}}(p_1p_2)\cdot\left(\frac{1}{\log\dfrac{x}{p_1p_2}}\right)\left(\frac{d\omega}{(p_1p_2)^\omega}\right)\right\}. \tag{6}$$

Let

$$M_1 = \sum_{(d_1,x)=1}\sum_{(d_2,x)=1}\frac{\lambda_{d_1}\lambda_{d_2}}{\varphi\left(\dfrac{d_1d_2}{(d_1,d_2)}\right)}\sum_{\substack{x^{\frac{1}{10}}<p_1\leqslant x^{\frac{1}{3}}<p_2\leqslant(\frac{x}{p_1})^{\frac{1}{2}}\\ n\leqslant\frac{x}{p_1p_2}}}\left(\frac{1}{\log\dfrac{x}{p_1p_2}}\right)\Lambda(n)\Phi\left(\frac{x}{p_1p_2n}\right),$$

$$M_2 = \sum_{\substack{d\leqslant x^{\frac{1}{2}-\epsilon}\\ (d,x)=1}}\frac{|\mu(d)|3^{\nu(d)}}{\varphi(d)}\left|\sideset{}{^*}\sum_{\chi_{d^*}\neq\chi_0}\overline{\chi^*_{d^*}}(x)\int_{2-i\infty}^{2+i\infty}\left(\frac{x^\omega}{\omega}\right)\left(1+\frac{\omega}{(\log x)^{1.1}}\right)^{-[\log x]-1}\right.$$

$$\left.\cdot\frac{L'}{L}(\omega,\chi^*_{d^*})\sum_{\substack{x^{\frac{1}{10}}<p_1\leqslant x^{\frac{1}{3}}<p_2\leqslant(\frac{x}{p_1})^{\frac{1}{2}}\\ (p_1p_2,d)=1}}\chi^*_{d^*}(p_1p_2)\left((p_1p_2)^\omega\log\frac{x}{p_1p_2}\right)^{-1}d\omega\right|.$$

Here d^* is the conductor of χ_d, and $\chi^*_{d^*}$ is the primitive character mod d^* equivalent to χ_d. $\nu(d)$ is the number of prime factors of d.

Lemma 5. *Let x be even; then*

$$\Omega \leqslant \frac{M_1+M_2}{1-\epsilon} + O\left(\frac{x}{(\log x)^{2.01}}\right).$$

Proof. By (5) and (6), we have

$$M \leqslant M_1 + |M_3| + M_4 + O\left(\frac{x}{(\log x)^{2.01}}\right), \tag{4}$$

where

$$M_3 = \sum_{(d_1,x)=1} \sum_{(d_2,x)=1} \frac{\lambda_{d_1}\lambda_{d_2}}{\varphi\left(\frac{d_1 d_2}{(d_1,d_2)}\right)} \sum_{\substack{x^{\frac{1}{10}}<p_1\leqslant x^{\frac{1}{3}}<p_2\leqslant(\frac{x}{p_1})^{\frac{1}{2}} \\ n\leqslant \frac{x}{p_1p_2} \\ (d_1d_2,p_1p_2n)>1}} \left(\frac{1}{\log\frac{x}{p_1p_2}}\right) \Lambda(n)\Phi\left(\frac{x}{p_1p_2n}\right),$$

$$M_4 = \sum_{(d_1,x)=1} \sum_{(d_2,x)=1} \left(-\frac{\lambda_{d_1}\lambda_{d_2}}{2\pi i\,\varphi\left(\frac{d_1 d_2}{(d_1,d_2)}\right)}\right) \int_{2-i\infty}^{2+i\infty} \left(\frac{x^\omega}{\omega}\right)\left(1+\frac{\omega}{(\log x)^{1.1}}\right)^{-[\log x]-1}$$

$$\cdot \sum_{\chi_{\frac{d_1d_2}{(d_1,d_2)}}\neq\chi_0} \overline{\chi_{\frac{d_1d_2}{(d_1,d_2)}}}(x)\frac{L'}{L}\left(\omega,\chi_{\frac{d_1d_2}{(d_1,d_2)}}\right) \sum_{x^{\frac{1}{10}}<p_1\leqslant x^{\frac{1}{3}}<p_2\leqslant(\frac{x}{p_1})^{\frac{1}{2}}} \frac{\chi_{\frac{d_1d_2}{(d_1,d_2)}}(p_1p_2)}{(p_1p_2)^\omega\log\frac{x}{p_1p_2}}\,d\omega.$$

We first estimate M_3:

$$M_3 \ll x^\epsilon \sum_{d\leqslant x^{\frac{1}{2}-\epsilon}} \frac{1}{d} \sum_{\substack{x^{\frac{1}{10}}<p_1\leqslant x^{\frac{1}{3}}<p_2\leqslant(\frac{x}{p_1})^{\frac{1}{2}} \\ n\leqslant\frac{x}{p_1p_2} \\ (d,p_1p_2n)>1}} \Lambda(n) \ll \sum_{x^{\frac{1}{10}}<p_1\leqslant x^{\frac{1}{3}}<p_2\leqslant(\frac{x}{p_1})^{\frac{1}{2}}} \left(\frac{x^{1+\epsilon}}{p_1p_2}\right)\left(\sum_{\substack{d\leqslant x^{\frac{1}{2}-\epsilon} \\ p_1|d}}\frac{1}{d}\right.$$

$$\left. + \sum_{\substack{d\leqslant x^{\frac{1}{2}-\epsilon} \\ p_2|d}}\frac{1}{d}\right) + \sum_{x^{\frac{1}{10}}<p_1\leqslant x^{\frac{1}{3}}<p_2\leqslant(\frac{x}{p_1})^{\frac{1}{2}}} \sum_{p\leqslant\frac{x}{p_1p_2}} (\log p) \sum_{\substack{d\leqslant x^{\frac{1}{2}-\epsilon} \\ p|d}} \frac{x^\epsilon}{d} + x^{1-\epsilon} \ll x^{1-\epsilon}. \tag{8}$$

We next estimate M_4. Let $\mu(d)\neq 0$, $d=p_1\cdots p_k$; then the necessary and sufficient condition for positive integers d_1, d_2 to satisfy $\frac{d_1d_2}{(d_1,d_2)} = d$ is $d_1 = p_1^{\alpha_1}\cdots p_k^{\alpha_k}$, $d_2 = p_1^{\beta_1}\cdots p_k^{\beta_k}$, where $0\leqslant\alpha_i\leqslant 1$, $0\leqslant\beta_i\leqslant 1$, $\alpha_i+\beta_i\geqslant 1$ $(1\leqslant i\leqslant k)$. Hence, when $d>0$ and $\mu(d)\neq 0$, the number of pairs d_1,d_2 satisfying $\frac{d_1d_2}{(d_1,d_2)} = d$ is $3^{\nu(d)}$. Since $|\lambda_d|\leqslant 1$, we have

$$M_4 \leqslant \sum_{\substack{d\leqslant x^{\frac{1}{2}-\epsilon} \\ (d,x)=1}} \frac{3^{\nu(d)}|\mu(d)|}{\varphi(d)}\Bigg|\sum_{\chi_d\neq\chi_0}^{*} \int_{2-i\infty}^{2+i\infty}\left(\frac{x^\omega}{\omega}\right)\left(1+\frac{\omega}{(\log x)^{1.1}}\right)^{-[\log x]-1}$$

$$\cdot \overline{\chi_d}(x) \frac{L'}{L}(\omega, \chi_d) \sum_{\substack{x^{\frac{1}{10}} < p_1 \leqslant x^{\frac{1}{3}} < p_2 \leqslant (\frac{x}{p_1})^{\frac{1}{2}} \\ (p_1p_2,d)=1}} \chi_d(p_1p_2) \left(\frac{d\omega}{(p_1p_2)^{\omega} \log \frac{x}{p_1p_2}} \right) \Big|.$$

Since $\frac{L'}{L}(\omega, \chi_d) = \frac{L'}{L}(\omega, \chi^*_{d^*}) + \sum_{p|\frac{d}{d^*}} \frac{\chi^*_{d^*}(p) \log p}{p^{\omega} - \chi^*_{d^*}(p)}$, we have

$$M_4 \leqslant M_2 + M_5, \tag{5}$$

where

$$M_5 = \sum_{\substack{d \leqslant x^{\frac{1}{2}-\epsilon} \\ (d,x)=1}} \frac{|\mu(d)| 3^{\nu(d)}}{\varphi(d)} \Big| \sideset{}{^*}\sum_{\chi_d \neq \chi_0} \overline{\chi^*_{d^*}}(x) \left(\sum_{p|\frac{d}{d^*}} \frac{\chi^*_{d^*}(p) \log p}{p^{\omega} - \chi^*_{d^*}(p)} \right) \sum_{\substack{x^{\frac{1}{10}} < p_1 \leqslant x^{\frac{1}{3}} < p_2 \leqslant (\frac{x}{p_1})^{\frac{1}{2}} \\ (p_1p_2,d)=1}} \frac{\chi^*_{d^*}(p_1p_2)}{(p_1p_2)^{\omega} \log \frac{x}{p_1p_2}} \, d\omega \Big|.$$

Also, when $\mathrm{Re}\,\omega = 2$, we have $\frac{\chi^*_{d^*}(p)}{p^{\omega} - \chi^*_{d^*}(p)} = \sum_{\lambda=1}^{\infty} \left(\frac{\chi^*_{d^*}(p)}{p^{\omega}} \right)^{\lambda}$. And when $\lambda \geqslant 1$, $\mu(d^*) \neq 0$, $(d^*, xp_1p_2p^{\lambda}) = 1$, applying Lemma 4 gives

$$\left| \sideset{}{^*}\sum_{\chi_{d^*}} \overline{\chi_{d^*}}(x) \chi_{d^*}(p_1p_2p^{\lambda}) \right| = \left| \sideset{}{^*}\sum_{\chi_{d^*}} \chi_{d^*}(p_1p_2p^{\lambda}y) \right| \leqslant |(p_1p_2p^{\lambda}y - 1, d^*)| = |(x - p_1p_2p^{\lambda}, d^*)|, \tag{6}$$

where y satisfies $xy \equiv 1 \pmod{d^*}$. From (10) and Lemma 1 we obtain

$$M_5 \ll \sum_{\substack{d \leqslant x^{\frac{1}{2}-\epsilon} \\ (d,x)=1}} \frac{|\mu(d)| 3^{\nu(d)}}{\varphi(d)} \Big| \sum_{\substack{d^*|d \\ d^*>1}} \sum_{p|\frac{d}{d^*}} (\log p) \sum_{\lambda=1}^{\infty} \sum_{\substack{x^{\frac{1}{10}} < p_1 \leqslant x^{\frac{1}{3}} < p_2 \leqslant (\frac{x}{p_1})^{\frac{1}{2}} \\ (p_1p_2,d)=1}}$$

$$\left(\frac{1}{\log \frac{x}{p_1p_2}} \right) \Phi \left(\frac{x}{p_1p_2p^{\lambda}} \right) \Bigg| \ll \sum_{\substack{d \leqslant x^{\frac{1}{2}-\epsilon} \\ (d,x)=1}} \frac{|\mu(d)| 3^{\nu(d)}}{\varphi(d)} \sum_{\substack{d^*|d \\ d^*>1}} \sum_{p|\frac{d}{d^*}} \sum_{x^{\frac{1}{10}} < p_1 \leqslant x^{\frac{1}{3}} < p_2 \leqslant (\frac{x}{p_1})^{\frac{1}{2}}}$$

$$\cdot \sum_{1 \leqslant \lambda \leqslant \left(\log \frac{x}{p_1p_2} \right) (\log p)^{-1}} \left(\frac{\log p}{\log \frac{x}{p_1p_2}} \right) (x - p_1p_2p^{\lambda}, d^*) \ll \sum_{\substack{k_1,k_2 \leqslant x^{\frac{1}{2}-\epsilon} \\ (k_1k_2,x)=1}} \frac{|\mu(k_1)||\mu(k_2)| x^{\frac{\epsilon}{4}}}{\varphi(k_1)\varphi(k_2)}$$

$$\cdot \sum_{p,k_2} \sum_{x^{\frac{1}{10}} < p_1 \leqslant x^{\frac{1}{3}} < p_2 \leqslant (\frac{x}{p_1})^{\frac{1}{2}}} \sum_{1 \leqslant \lambda \leqslant \left(\log \frac{x}{p_1p_2} \right) (\log p)^{-1}} (x - p_1p_2p^{\lambda}, k_1) \ll x^{\frac{\epsilon}{3}} \sum_{\substack{k_1 \leqslant x^{\frac{1}{2}-\epsilon} \\ (k_1,x)=1}} \frac{1}{k_1}$$

$$\cdot \sum_{\substack{x^{\frac{1}{10}} < p_1 \leqslant x^{\frac{1}{3}} < p_2 \leqslant (\frac{x}{p_1})^{\frac{1}{2}} \\ p^{\lambda} \leqslant \frac{x}{p_1p_2}}} (x - p_1p_2p^{\lambda}, k_1) \sum_{\substack{k_2 \leqslant x^{\frac{1}{2}-\epsilon} \\ k_2 \equiv 0 \ (\bmod p)}} \frac{1}{k_2} \ll x^{\frac{\epsilon}{2}} \sum_{\substack{x^{\frac{1}{10}} < p_1 \leqslant x^{\frac{1}{3}} < p_2 \leqslant (\frac{x}{p_1})^{\frac{1}{2}} \\ p^{\lambda} \leqslant \frac{x}{p_1p_2}}}$$

$$\cdot \frac{1}{p} \sum_{d|(x - p_1p_2p^{\lambda})} d \sum_{\substack{k_1 \leqslant x^{\frac{1}{2}-\epsilon} \\ d|k_1}} \frac{1}{k_1} \ll x^{1-\epsilon}. \tag{11}$$

By (7), (8), (9), and (11), the lemma is proved. □

Lemma 6. *We have*

$$M_2 \ll \frac{x}{(\log x)^{2.01}}.$$

Proof. Let

$$\Phi(y,\chi) = \int_{2-i\infty}^{2+i\infty} \left(\frac{y^\omega}{\omega}\right)\left(1+\frac{\omega}{(\log x)^{1.1}}\right)^{-[\log x]-1} \frac{L'}{L}(\omega,\chi)\,d\omega = \int_{1+\frac{1}{\log x}-i\infty}^{1+\frac{1}{\log x}+i\infty} \left(\frac{y^\omega}{\omega}\right)\left(1+\frac{\omega}{(\log x)^{1.1}}\right)^{-[\log x]-1}$$

Then

$$M_2 \leqslant \sum_{\substack{1<l\leqslant x^{\frac{1}{2}-\epsilon}\\(l,x)=1}} \left\{\sum_{\substack{1<d\leqslant x^{\frac{1}{2}-\epsilon}\\ l|d,\,(d,x)=1}} \frac{|\mu(d)|3^{\nu(d)}}{\varphi(d)}\right\} \Big|\sideset{}{^*}\sum_{\chi_l}\overline{\chi_l}(x) \sum_{\substack{x^{\frac{1}{10}}<p_1\leqslant x^{\frac{1}{3}}<p_2\leqslant(\frac{x}{p_1})^{\frac{1}{2}}\\(p_1p_2,d)=1}} \left(\frac{1}{\log\dfrac{x}{p_1p_2}}\right)$$

$$\cdot\,\Phi\left(\frac{x}{p_1p_2},\chi_l\right)\chi_l(p_1p_2)\Big| \leqslant \sum_{\substack{1<d\leqslant x^{\frac{1}{2}-\epsilon}\\(d,x)=1}} \frac{|\mu(d)|3^{\nu(d)}}{\varphi(d)}$$

$$\cdot\left\{\sum_{\substack{1<l\leqslant x^{\frac{1}{2}-\epsilon}\\(l,xd)=1}} \frac{|\mu(l)|3^{\nu(l)}}{\varphi(l)} \Big|\sideset{}{^*}\sum_{\chi_l}\overline{\chi_l}(x) \sum_{\substack{x^{\frac{1}{10}}<p_1\leqslant x^{\frac{1}{3}}<p_2\leqslant(\frac{x}{p_1})^{\frac{1}{2}}\\(p_1p_2,d)=1}} \left(\frac{1}{\log\dfrac{x}{p_1p_2}}\right)\Phi\left(\frac{x}{p_1p_2},\chi_l\right)\chi_l(p_1p_2)\Big|\right\}.$$

Let $\tau(l) = \sum_{d|l} 1$; then

$$\sum_{1<d\leqslant x^{\frac{1}{2}-\epsilon}} \frac{3^{\nu(d)}|\mu(d)|}{\varphi(d)} \ll (\log x)\sum_{d\leqslant x^{\frac{1}{2}-\epsilon}}\frac{(\tau(d))^2}{d} \ll (\log x)^5.$$

Hence

$$M_2 \ll (\log x)^6 \max_{1<m\leqslant x^{\frac{1}{2}}} N_m. \tag{7}$$

where

$$N_m = \sum_{\substack{1<l\leqslant x^{\frac{1}{2}-\epsilon}\\(l,x)=1}} \frac{|\mu(l)|3^{\nu(l)}}{l} \Big|\sideset{}{^*}\sum_{\chi_l}\overline{\chi_l}(x) \sum_{\substack{x^{\frac{1}{10}}<p_1\leqslant x^{\frac{1}{3}}<p_2\leqslant(\frac{x}{p_1})^{\frac{1}{2}}\\(p_1p_2,m)=1}} \left(\frac{1}{\log\dfrac{x}{p_1p_2}}\right)\Phi\left(\frac{x}{p_1p_2},\chi_l\right)\chi_l(p_1p_2)\Big|.$$

We write $\sum_{(k,m)}$ for a sum in which p_1 and p_2 range over exactly $x^{\frac{1}{10}} < p_1 \leqslant x^{\frac{1}{3}} < p_2 \leqslant \left(\frac{x}{p_1}\right)^{\frac{1}{2}}$, $x^{\frac{13}{30}}2^k < p_1p_2 \leqslant x^{\frac{13}{30}}2^{k+1}$, $(p_1p_2,m)=1$. Let I_1 be a positive integer satisfying $2^{I_1-1}(\log x)^{100} < x^{\frac{1}{2}-\epsilon} < 2^{I_1}(\log x)^{100}$, $I_2 = \left[\dfrac{7\log x}{30\log 2}\right]$; then

$$N_m \leqslant \sum_{l=0}^{I_1}\sum_{k=0}^{I_2} N_m^{(l,k)}. \tag{8}$$

where

$$N_m^{(0,k)} = \sum_{\substack{1<d\leqslant(\log x)^{100}\\(d,x)=1}} \frac{|\mu(d)|3^{\nu(d)}}{d} \Big|\sideset{}{^*}\sum_{\chi_d} \overline{\chi_d}(x) \sum_{(k,m)} \left(\frac{1}{\log\dfrac{x}{p_1p_2}}\right) \Phi\left(\frac{x}{p_1p_2}, \chi_d\right) \chi_d(p_1p_2)\Big|.$$

and when $l \geqslant 1$,

$$N_m^{(l,k)} = \sum_{\substack{2^{l-1}(\log x)^{100}<d\leqslant 2^l(\log x)^{100}\\(d,x)=1}} \frac{|\mu(d)|3^{\nu(d)}}{d} \Big|\sideset{}{^*}\sum_{\chi_d} \overline{\chi_d}(x) \sum_{(k,m)} \left(\frac{1}{\log\dfrac{x}{p_1p_2}}\right) \Phi\left(\frac{x}{p_1p_2}, \chi_d\right) \chi_d(p_1p_2)\Big|.$$

Let $S(H,\omega,\chi_d) = \sum_{n=1}^{H} \frac{\mu(n)\chi_d(n)}{n^\omega}$, where $H \ll x$. We know that when $\operatorname{Re}\omega \geqslant 1$,

$$S(H,\omega,\chi_d) \ll \log x; \qquad L(\omega,\chi_d) = \sum_{n=1}^{H} \frac{\chi_d(n)}{n^\omega} + O\left(\frac{|\omega| d^{\frac{1}{2}} \log d}{H}\right).$$

Hence, when $\operatorname{Re}\omega \geqslant 1$,

$$1 - L(\omega,\chi_d)S(H,\omega,\chi_d) = \sum_{n=1}^{\infty} \frac{C_H(n)\chi_d(n)}{n^\omega} + O\left(\frac{|\omega| d^{\frac{1}{2}} (\log x)^2}{H}\right),$$

where $C_H(1) = 0$; when $n > H^2$, $C_H(n) = 0$; and when $n > 1$, $C_H(n) = -\sum_d \mu(d)$, where d ranges over the divisors of n such that $1 \leqslant d \leqslant H$ and $\frac{n}{d} \leqslant H$; when $1 \leqslant n \leqslant H$, $C_H(n) = 0$; and when $n > H$, $C_H(n) \leqslant \tau(n)$. Hence, for $H \ll x$, by the Schwarz inequality we obtain

$$\left|\sum_{n=1}^{\infty} \frac{C_H(n)\chi_d(n)}{n^\omega}\right|^2 \ll (\log x) \sum_{l=0}^{3I_1} \left|\sum_{n=2^lH+1}^{2^{l+1}H} \frac{C_H(n)\chi_d(n)}{n^\omega}\right|^2.$$

Let $\alpha = 1 + \frac{1}{\log x}$. From the above, $\sum_{n\leqslant x} \tau^2(n) \ll x(\log x)^3$, and (3), we obtain: when $Q \ll x$,

$$\sum_{D<d\leqslant Q} \frac{1}{\varphi(d)} \sideset{}{^*}\sum_{\chi_d} \left|\sum_{n=2^lH+1}^{2^{l+1}H} \frac{C_H(n)\chi_d(n)}{n^{\alpha+i\nu}}\right|^2 \ll \left(Q + \frac{2^lH}{D}\right) \sum_{n=2^lH+1}^{2^{l+1}H} \frac{(\tau(n))^2}{n^2} \ll \left(\frac{Q}{2^lH} + \frac{1}{D}\right)(\log x)^3,$$

and

$$\sum_{D<d\leqslant Q} \frac{1}{\varphi(d)} \sideset{}{^*}\sum_{\chi_d} |1 - L(\alpha+i\nu,\chi_d)S(H,\alpha+i\nu,\chi_d)|^2 \ll \sum_{D<d\leqslant Q} \frac{1}{\varphi(d)} \sideset{}{^*}\sum_{\chi_d} \left|\sum_{n=1}^{\infty} \frac{C_H(n)\chi_d(n)}{n^{\alpha+i\nu}}\right|^2 + \frac{|\alpha+i\nu|^2Q^2(\log}{H^2}$$

$$\ll \left(\frac{Q}{H} + \frac{1}{D} + \frac{|\alpha+i\nu|^2Q^2}{H^2}\right)(\log x)^5. \tag{9}$$

Let $\beta = \dfrac{1}{2} + \dfrac{1}{\log x}$. Since $\{S(H, \beta + i\nu, \chi_d)\}^2 = \displaystyle\sum_{n=1}^{H^2} \frac{j(n)\chi_d(n)}{n^{\beta+i\nu}}$, where $|j(n)| \leqslant \tau(n)$, it follows from (3) that, when $l \geqslant 1$, $H \ll x$,

$$\sum_{2^{l-1}(\log x)^{100} < d \leqslant 2^l (\log x)^{100}} \frac{1}{\varphi(d)} \sideset{}{^*}\sum_{\chi_d} |S(H, \beta + i\nu, \chi_d)|^4 \ll \left(2^l (\log x)^{100} + \frac{H^2}{2^l (\log x)^{100}} \right) \sum_{n=1}^{H^2} \frac{(\tau(n))^2}{n} \ll 2^l (\log x \tag{10}$$

Since $L'(\omega, \chi_d) = \dfrac{1}{2\pi i} \displaystyle\int_r \frac{L(\xi, \chi_d)}{(\xi - \omega)^2}\, d\xi$, where r is the circle of radius $(\log x)^{-1}$ centered at ω, we have

$$|L'(\omega, \chi_d)| \ll (\log x)^2 \int_r |L(\xi, \chi_d)|\, d\xi.$$

By Hölder's inequality,

$$|L'(\omega, \chi_d)|^4 \ll (\log x)^5 \int_r |L(\xi, \chi_d)|^4\, |d\xi|.$$

And by Lemma 3, we have

$$\sum_{2^{l-1}(\log x)^{100} < d \leqslant 2^l (\log x)^{100}} \left(\frac{1}{\varphi(d)} \right) \sideset{}{^*}\sum_{\chi_d} |L'(\beta + i\nu, \chi_d)|^4 \ll 2^l (\log x)^{109} (|\beta + i\nu|)^2.$$

When $\mathrm{Re}\, \omega \geqslant \alpha = 1 + \dfrac{1}{\log x}$, we obtain

$$\frac{L'}{L}(\omega, \chi_d) = \left\{ \frac{L'}{L}(\omega, \chi_d) \right\} \{1 - L(\omega, \chi_d) S(H, \omega, \chi_d)\} + L'(\omega, \chi_d) S(H, \omega, \chi_d). \tag{11}$$

Let

$$A(l, k, \omega, m, H) = \sum_{\substack{2^{l-1}(\log x)^{100} < d \leqslant 2^l (\log x)^{100} \\ (d,x)=1}} \frac{|\mu(d)| 3^{\nu(d)}}{d} \sideset{}{^*}\sum_{\chi_d} \left| \sum_{(k,m)} \frac{\chi_d(p_1 p_2)}{(p_1 p_2)^{\omega} \log \dfrac{x}{p_1 p_2}} \right| |1 - L(\omega, \chi_d) S(H, \omega, \chi_d)|,$$

$$B(l, k, \omega, m, H) = \sum_{\substack{2^{l-1}(\log x)^{100} < d \leqslant 2^l (\log x)^{100} \\ (d,x)=1}} \frac{|\mu(d)| 3^{\nu(d)}}{d} \sideset{}{^*}\sum_{\chi_d} \left| \sum_{(k,m)} \frac{\chi_d(p_1 p_2)}{(p_1 p_2)^{\omega} \log \dfrac{x}{p_1 p_2}} \right| |L'(\omega, \chi_d) S(H, \omega, \chi_d)|.$$

For $l \geqslant 1$, (16) gives

$$N_m^{(l,k)} \ll x (\log x)^2 \int_0^{\infty} \frac{A(l, k, \alpha + i\nu, m, H)}{|\alpha + i\nu| \left(1 + \dfrac{|\alpha + i\nu|}{(\log x)^{1.1}} \right)^{[\log x]+1}}\, d\nu + x^{\frac{3}{2}} \int_0^{\infty} \frac{B(l, k, \beta + i\nu, m, H)}{|\beta + i\nu| \left(1 + \dfrac{|\beta + i\nu|}{(\log x)^{1.1}} \right)^{[\log x]+1}}\, d\nu. \tag{12}$$

It is clear that, when $|\mu(d)| \neq 0$ and d is large,

$$3^{\nu(d)} \leqslant e^{\frac{3 \log d}{\log \log d}}. \tag{13}$$

We first estimate $N_m^{(l,k)}$ for $l \geqslant 1$ in the two cases $2^k x^{\frac{13}{30}} > x^{\frac{1}{2}-\epsilon}$ and $x^{\frac{1}{2}-\epsilon} \geqslant 2^k x^{\frac{13}{30}} > 2^l (\log x)^{100}$; here we take $H = 2^l (\log x)^{200} I_{l,x}$, where $I_{l,x} = e^{\frac{6\log\{2^l(\log x)^{100}\}}{\log\log\{2^l(\log x)^{100}\}}}$. Then, by (14)–(18),

$$
\begin{aligned}
N_m^{(l,k)} \ll\ & x(\log x)^4 \int_0^\infty \Bigg[\Bigg\{ \sum_{\substack{2^{l-1}(\log x)^{100} < d \leqslant 2^l (\log x)^{100} \\ (d,x)=1}} \frac{|\mu(d)|}{d} \sum_{\chi_d}^{*} \left| \sum_{(k,m)} \frac{\chi_d(p_1p_2)}{(p_1p_2)^{\alpha+i\nu} \log \frac{x}{p_1p_2}} \right|^2 \Bigg\} \\
& \cdot \Bigg\{ \sum_{\substack{2^{l-1}(\log x)^{100} < d \leqslant 2^l (\log x)^{100} \\ (d,x)=1}} \frac{|\mu(d)|}{d} \sum_{\chi_d}^{*} \left|1 - L(\alpha+i\nu, \chi_d) S(H, \alpha+i\nu, \chi_d)\right|^2 I_{l,x} \Bigg\}^{\frac{1}{2}} \Bigg] \left(\frac{d\nu}{1+\nu^{2.1}} \right) \\
& + x^{\frac{3}{2}} (\log x)^4 \int_0^\infty \Bigg\{ (I_{l,x}) \sum_{\substack{2^{l-1}(\log x)^{100} < d \leqslant 2^l (\log x)^{100} \\ (d,x)=1}} \frac{|\mu(d)|}{d} \sum_{\chi_d}^{*} \left| \sum_{(k,m)} \frac{\chi_d(p_1p_2)}{(p_1p_2)^{\beta+i\nu} \log \frac{x}{p_1p_2}} \right|^2 \Bigg\}^{\frac{1}{2}} \\
& \cdot \Bigg\{ \sum_{\substack{2^{l-1}(\log x)^{100} < d \leqslant 2^l (\log x)^{100} \\ (d,x)=1}} \frac{|\mu(d)|}{d} \sum_{\chi_d}^{*} |S(H, \beta+i\nu, \chi_d)|^4 \Bigg\}^{\frac{1}{4}} \left(\frac{d\nu}{1+\nu^4} \right) \Bigg\{ \sum_{\substack{2^{l-1}(\log x)^{100} < d \leqslant 2^l (\log x)^{100} \\ (d,x)=1}} |\mu \\
\ll\ & x(\log x)^8 \int_0^\infty \Bigg\{ \left(2^l (\log x)^{100} + \frac{2^k x^{\frac{13}{30}}}{2^l (\log x)^{100}} \right) \Bigg(\sum_{2^k x^{\frac{13}{30}} < n \leqslant 2^{k+1} x^{\frac{13}{30}}} \frac{1}{n^2} \Bigg) \Bigg(\frac{2^l (\log x)^{100}}{H} \\
& + \frac{1}{2^l (\log x)^{100}} + \frac{(1+\nu^2) 2^{2l} (\log x)^{200}}{H^2} \Bigg) (I_{l,x}) \Bigg\}^{\frac{1}{2}} \left(\frac{d\nu}{1+\nu^{2.1}} \right) + x^{\frac{3}{2}} (\log x)^8 \int_0^\infty \Bigg\{ \Big(2^l (\log x)^{100} \\
& + \frac{2^k x^{\frac{13}{30}}}{2^l (\log x)^{100}} \Big) (I_{l,x}) \Bigg\}^{\frac{1}{2}} \left\{ 2^{2l} (\log x)^{213} + H^2 (\log x)^{13} \right\}^{\frac{1}{4}} (1+\nu^2)^{\frac{1}{4}} \left(\frac{d\nu}{1+\nu^4} \right) \ll \frac{x}{(\log x)^{20}}.
\end{aligned}
\tag{19}
$$

We now estimate $N_m^{(l,k)}$ for $2^k x^{\frac{13}{30}} \leqslant 2^l (\log x)^{100} \leqslant 2x^{\frac{1}{2}-\epsilon}$; here we take

$$
H = \max\left(2^{2l-k} x^{-\frac{13}{30}} (\log x)^{400} I_{l,x},\ x^{\frac{1}{2}-\epsilon} \right),
$$

then

$$N_m^{(l,k)} \ll x(\log x)^8 \int_0^\infty \left\{ \left(2^l (\log x)^{100} + \frac{2^k x^{\frac{13}{30}}}{2^l (\log x)^{100}} \right) \left(\sum_{2^k x^{\frac{13}{30}} < n \leqslant 2^{k+1} x^{\frac{13}{30}}} \frac{1}{n^2} \right) \right.$$

$$\left. \cdot \left(\frac{2^l (\log x)^{100}}{H} + \frac{1}{2^l (\log x)^{100}} + \frac{(1+\nu^2) 2^{2l} (\log x)^{200}}{H^2} \right) (I_{l,x}) \right\}^{\frac{1}{2}} \left(\frac{d\nu}{1+\nu^{2.1}} \right)$$

$$+ x^{\frac{1}{2}} (\log x)^4 \int_0^\infty \left\{ \sum_{\substack{2^{l-1}(\log x)^{100} < d \leqslant 2^l (\log x)^{100} \\ (d,x)=1}} \frac{|\mu(d)|}{d} \sideset{}{^*}\sum_{\chi_d} |S(H, \beta + i\nu, \chi_d)|^2 \right\}^{\frac{1}{2}} (I_{l,x})^{\frac{1}{2}}$$

$$\cdot \left\{ \sum_{\substack{2^{l-1}(\log x)^{100} < d \leqslant 2^l (\log x)^{100} \\ (d,x)=1}} \frac{|\mu(d)|}{d} \sideset{}{^*}\sum_{\chi_d} \left| \left(\sum_{(k,m)} \frac{\chi_d(p_1 p_2)}{(p_1 p_2)^{\beta + i\nu} \log \frac{x}{p_1 p_2}} \right)^2 \right|^2 \right\}^{\frac{1}{4}} \left(\frac{d\nu}{1+\nu^4} \right)$$

$$\ll \frac{x}{(\log x)^{20}} + x^{\frac{1}{2}} (\log x)^{20} \left\{ 2^l (\log x)^{100} + \frac{H}{2^l (\log x)^{100}} \right\}^{\frac{1}{2}} (I_{l,x})^{\frac{3}{2}} \left(2^l (\log x)^{100} \right)^{\frac{1}{4}}$$

$$\cdot \left(2^l (\log x)^{100} + \frac{2^{2k} x^{\frac{13}{15}}}{2^l (\log x)^{100}} \right)^{\frac{1}{4}} \int_0^\infty \frac{(1+\nu^2)^{\frac{1}{4}}}{1+\nu^4} \, d\nu \ll \frac{x}{(\log x)^{20}}. \tag{20}$$

We now estimate $N_m^{(0,k)}$, for $0 \leqslant k \leqslant I_2$. When χ_d is a primitive character and $\operatorname{Re} S \geqslant 1 - \frac{c}{d^{\frac{1}{300}}}$, we have $L(S, \chi_d) \neq 0$, where c is a constant. Hence

$$N_m^{(0,k)} \ll \sum_{1 < d \leqslant (\log x)^{100}} \frac{3^{\nu(d)} |\mu(d)|}{d} \sideset{}{^*}\sum_{\chi_d} \left| \int_{1 - \frac{1}{(\log x)^{1/2}} - i\infty}^{1 - \frac{1}{(\log x)^{1/2}} + i\infty} \sum_{(k,m)} \left(\frac{1}{\log \frac{x}{p_1 p_2}} \right) \right.$$

$$\left. \cdot \chi_d(p_1 p_2) \left(\frac{x}{p_1 p_2} \right)^\omega \left(1 + \frac{\omega}{(\log x)^{1.1}} \right)^{-[\log x] - 1} \frac{L'}{L}(\omega, \chi_d) \, d\omega \right|$$

$$\ll (\log x)^{200} \sum_{x^{\frac{1}{10}} < p_1 \leqslant x^{\frac{1}{3}} < p_2 \leqslant \left(\frac{x}{p_1}\right)^{\frac{1}{2}}} \left(\frac{x}{p_1 p_2} \right)^{1 - \frac{1}{(\log x)^{1/2}}} \ll \frac{x}{(\log x)^{20}}. \tag{21}$$

By (12), (13), and (19)–(21), the lemma is proved. $\square$

Lemma 7. *For large even x, we have*

$$M_1 \leqslant \left\{ \frac{(8 + 24\epsilon) x C_x}{\log x} \right\} \left\{ \sum_{x^{\frac{1}{10}} < p_1 \leqslant x^{\frac{1}{3}} < p_2 \leqslant \left(\frac{x}{p_1}\right)^{\frac{1}{2}}} \frac{1}{p_1 p_2 \log \frac{x}{p_1 p_2}} \right\},$$

where $C_x = \prod_{\substack{p \mid x \\ p > 2}} \frac{p-1}{p-2} \prod_{p > 2} \left(1 - \frac{1}{(p-1)^2} \right)$.

Proof. Let $S = \sum_{\substack{1 \leqslant k \leqslant (x^{\frac{1}{2}-\epsilon})^{\frac{1}{2}} \\ (k,x)=1}} \frac{\mu^2(k)}{f(k)}$; then

$$\lambda_d g(d) = \left(\frac{1}{S}\right) \sum_{\substack{1 \leqslant k \leqslant (x^{\frac{1}{2}-\epsilon})^{\frac{1}{2}}/d \\ (k,xd)=1}} \frac{\mu(kd)\mu(k)}{f(kd)}.$$

When $(m,x)=1$, we have

$$\sum_{\substack{d \leqslant (x^{\frac{1}{2}-\epsilon})^{\frac{1}{2}} \\ (d,x)=1,\, m|d}} \lambda_d g(d) = \left(\frac{1}{S}\right) \left(\sum_{\substack{d \leqslant (x^{\frac{1}{2}-\epsilon})^{\frac{1}{2}} \\ (d,x)=1,\, m|d}} \sum_{\substack{1 \leqslant k \leqslant (x^{\frac{1}{2}-\epsilon})^{\frac{1}{2}}/d \\ (k,xd)=1}} \frac{\mu(kd)\mu(k)}{f(kd)} \right) = \left(\frac{1}{S}\right) \sum_{\substack{1 \leqslant r \leqslant (x^{\frac{1}{2}-\epsilon})^{\frac{1}{2}} \\ (r,x)=1}} \frac{\mu(r)}{f(r)} \sum_{m|d|r} \mu\left(\frac{r}{d}\right) = \frac{\mu}{S}$$

Since $\frac{1}{\varphi\left(\frac{d_1 d_2}{(d_1,d_2)}\right)} = g(d_1)g(d_2) \sum_{d|(d_1,d_2)} f(d)$, we have

$$\sum_{\substack{d_1 \leqslant (x^{\frac{1}{2}-\epsilon})^{\frac{1}{2}} \\ (d_1 d_2,x)=1}} \sum_{d_2 \leqslant (x^{\frac{1}{2}-\epsilon})^{\frac{1}{2}}} \frac{\lambda_{d_1}\lambda_{d_2}}{\varphi\left(\frac{d_1 d_2}{(d_1,d_2)}\right)} = \sum_{d_1 \leqslant (x^{\frac{1}{2}-\epsilon})^{\frac{1}{2}}} \sum_{d_2 \leqslant (x^{\frac{1}{2}-\epsilon})^{\frac{1}{2}}} \lambda_{d_1}\lambda_{d_2} g(d_1)g(d_2) \sum_{k|(d_1,d_2)} f(k)$$

$$= \sum_{\substack{k \leqslant (x^{\frac{1}{2}-\epsilon})^{\frac{1}{2}} \\ (k,x)=1}} f(k) \left(\sum_{\substack{d \leqslant (x^{\frac{1}{2}-\epsilon})^{\frac{1}{2}} \\ k|d,\,(d,x)=1}} \lambda_d g(d) \right)^2 = \frac{1}{S}. \qquad (22)$$

Let $V_k(x) = \sum_{\substack{1 \leqslant n \leqslant x \\ (n,k)=1}} \frac{\mu^2(n)}{\varphi(n)}$; then

$$\log x \leqslant \sum_{n=1}^{x} \frac{1}{n} \leqslant \sum_{1 \leqslant n \leqslant x} \frac{\mu^2(n)}{n} \prod_{p|n} \left(\sum_{l=0}^{\infty} \frac{1}{p^l} \right) = \sum_{1 \leqslant n \leqslant x} \frac{\mu^2(n)}{n} \prod_{p|n} \left(1 - \frac{1}{p}\right)^{-1} = V_1(x) = \sum_{d|k} \sum_{\substack{1 \leqslant n \leqslant x \\ (n,k)=d}} \frac{\mu^2(n)}{\varphi(n)}$$

$$= \sum_{d|k} \frac{\mu^2(d)}{\varphi(d)} \sum_{\substack{1 \leqslant m \leqslant x/d \\ (m,k)=1}} \frac{\mu^2(m)}{\varphi(m)} \leqslant \sum_{d|k} \frac{\mu^2(d)}{\varphi(d)} V_k(x) = \frac{k V_k(x)}{\varphi(k)},$$

hence $V_k(x) \geqslant \frac{\varphi(k)\log x}{k}$. Let $\phi(1)=1$, and for $q>2$ let $\phi(q) = \prod_{p|q}(p-2)$; then

$$S = \sum_{\substack{1 \leqslant k \leqslant (x^{\frac{1}{2}-\epsilon})^{\frac{1}{2}} \\ (k,x)=1}} \frac{\mu^2(k)}{\varphi(k)} \prod_{p|k} \left(1 + \frac{1}{p-2}\right) = \sum_{\substack{q \leqslant (x^{\frac{1}{2}-\epsilon})^{\frac{1}{2}} \\ (q,x)=1}} \frac{\mu^2(q)}{\phi(q)\varphi(q)} \sum_{\substack{1 \leqslant r \leqslant (x^{\frac{1}{2}-\epsilon})^{\frac{1}{2}}/q \\ (r,qx)=1}} \frac{\mu^2(r)}{\varphi(r)}$$

$$\geqslant \sum_{\substack{q \leqslant (x^{\frac{1}{2}-\epsilon})^{\frac{1}{2}} \\ (q,x)=1}} \frac{\mu^2(q)}{\phi(q)\varphi(q)} \left\{ \frac{\varphi(qx)}{qx} \log \frac{x^{\frac{1}{4}-\frac{\epsilon}{2}}}{q} \right\} = \left(\frac{\varphi(x)}{x}\right) \left(\log x^{\frac{1}{4}-\frac{\epsilon}{2}}\right) \prod_{p \nmid x} \left(1 + \frac{1}{p(p-2)}\right) + O(1) = \frac{\left(\frac{1}{8} - \right.}{}$$

From (22) and the above, for large x,

$$M_1 \leqslant (8+24\epsilon)C_x(\log x)^{-1} \sum_{\substack{x^{\frac{1}{10}}<p_1\leqslant x^{\frac{1}{3}}<p_2\leqslant(\frac{x}{p_1})^{\frac{1}{2}} \\ n\leqslant \frac{x}{p_1p_2}}} \left(\frac{\Lambda(n)}{\log \dfrac{x}{p_1p_2}}\right) \Phi\left(\frac{x}{p_1p_2n}\right).$$

By Lemma 1, the lemma is proved. $\square$

Lemma 8. *For large even x, we have*

$$\Omega \leqslant \frac{3.9404xC_x}{(\log x)^2}.$$

Proof. For large x, by Lemmas 5 through 7, we have

$$\Omega \leqslant \left\{\frac{8(1+5\epsilon)xC_x}{\log x}\right\}\left\{\sum_{x^{\frac{1}{10}}<p_1\leqslant x^{\frac{1}{3}}<p_2\leqslant(\frac{x}{p_1})^{\frac{1}{2}}} \frac{1}{p_1p_2\log\dfrac{x}{p_1p_2}}\right\}, \tag{14}$$

and also

$$\sum_{x^{\frac{1}{10}}<p_1\leqslant x^{\frac{1}{3}}<p_2\leqslant(\frac{x}{p_1})^{\frac{1}{2}}} \frac{1}{p_1p_2\log\dfrac{x}{p_1p_2}} \leqslant (1+\epsilon)\sum_{x^{\frac{1}{10}}<p_1\leqslant x^{\frac{1}{3}}} \int_{x^{\frac{1}{3}}}^{(\frac{x}{p_1})^{\frac{1}{2}}} \frac{dt}{p_1t(\log t)\log\dfrac{x}{p_1t}}$$

$$\leqslant (1+2\epsilon)\int_{x^{\frac{1}{10}}}^{x^{\frac{1}{3}}} \frac{dS}{S\log S}\int_{S^{\frac{1}{3}}}^{(\frac{x}{S})^{\frac{1}{2}}} \frac{dt}{t(\log t)\left(\log\dfrac{x}{St}\right)} = (1+2\epsilon)\int_{\frac{1}{10}}^{\frac{1}{3}} \frac{d\alpha}{\alpha}\int_{\frac{1}{3}}^{\frac{1-\alpha}{2}} \frac{d\beta}{\beta(1-\alpha-\beta)\log x},$$

and

$$\int_{\frac{1}{10}}^{\frac{1}{3}} \frac{d\alpha}{\alpha}\int_{\frac{1}{3}}^{\frac{1-\alpha}{2}} \left(\frac{1}{1-\alpha}\right)\left(\frac{1}{\beta}+\frac{1}{1-\alpha-\beta}\right)d\beta = \int_{\frac{1}{10}}^{\frac{1}{3}} \frac{\log\dfrac{1-\alpha}{2}-\log\dfrac{1}{3}-\log\dfrac{1-\alpha}{2}+\log\left(\dfrac{2}{3}-\alpha\right)}{\alpha(1-\alpha)}d\alpha$$

$$= \int_{\frac{1}{10}}^{\frac{1}{3}} \frac{\log(2-3\alpha)}{\alpha(1-\alpha)}d\alpha = \sum_{i=0}^{6}\int_{\frac{1}{10}+\frac{i}{30}}^{\frac{1}{10}+\frac{i+1}{30}} \frac{\log\left(1.6-\dfrac{i}{10}\right)}{\alpha(1-\alpha)}d\alpha + \sum_{i=0}^{6}\int_{\frac{1}{10}+\frac{i}{30}}^{\frac{1}{10}+\frac{i+1}{30}} \frac{\log\dfrac{2-3\alpha}{1.6-0.i}}{\alpha(1-\alpha)}d\alpha$$

$$\leqslant \sum_{i=0}^{6}\left\{\log\left(1.6-\frac{i}{10}\right)\right\}\left\{\log\frac{\dfrac{9}{10}-\dfrac{i}{30}}{\dfrac{1}{10}+\dfrac{i}{30}} - \log\frac{\dfrac{9}{10}-\dfrac{1}{30}-\dfrac{i}{30}}{\dfrac{1}{10}+\dfrac{i+1}{30}}\right\}$$

$$+\sum_{i=0}^{6}\int_{\frac{1}{10}+\frac{i}{30}}^{\frac{1}{10}+\frac{i+1}{30}} \frac{(0.4+0.i-3\alpha)}{(1.6-0.i)\alpha(1-\alpha)}d\alpha \leqslant \sum_{i=0}^{6}\left\{\log(1.6-0.i)+\frac{4+i}{16-i}\right\}\left\{\log\frac{27-i}{3+i}\right.$$

$$\left.-\log\frac{26-i}{4+i}\right\} - 3\sum_{i=0}^{6}\int_{\frac{1}{10}+\frac{i}{30}}^{\frac{1}{10}+\frac{i+1}{30}} \frac{d\alpha}{(1.6-0.i)(1-\alpha)} = \sum_{i=0}^{6}\left\{\log(1.6-0.i)+\frac{4+i}{16-i}\right\}$$

$$\cdot \left\{ \log \frac{108+23i-i^2}{78+23i-i^2} \right\} - 3\sum_{i=0}^{6} \left(\frac{1}{1.6-0.i} \right) \left(\log \frac{27-i}{26-i} \right)$$

$$\leqslant (0.47+0.25)(0.32542)+(0.40547+0.33334)(0.26236)+(0.33647+0.42858)(0.22315)+(0.26236+0.53847$$

$$+(0.18232+0.66667)(0.17799)+(0.09531+0.81819)(0.16431)+0.15415-3\left(\frac{0.03774}{1.6}+\frac{0.03922}{1.5}+\frac{0.04082}{1.4}\right.$$

$$\left.+\frac{0.04445}{1.2}+\frac{0.04652}{1.1}+0.04879\right) \leqslant 0.234303+0.193837+0.17073+0.15754+0.151115+0.1501+0.15415$$

$$-3(0.023587+0.026146+0.029157+0.032738+0.037041+0.04229+0.04879) \leqslant 1.21178-0.71924 = 0.4925 \tag{15}$$

By (23) and (24), Lemma 8 is proved. □

Let x be a large even integer, and let $P_x(x, x^{\frac{1}{10}})$ denote the number of primes p satisfying: $p \leqslant x$, $p \not\equiv x \pmod{p_i}$ $(1 \leqslant i \leqslant j)$, where $3 = p_1 < p_2 < \cdots < p_j \leqslant x^{\frac{1}{10}}$. For a prime p', let $P_x(x, p', x^{\frac{1}{10}})$ denote the number of primes p satisfying: $p \leqslant x$, $p \equiv x \pmod{p'}$, $p \not\equiv x \pmod{p_i}$ $(1 \leqslant i \leqslant j)$, where $3 = p_1 < p_2 < \cdots < p_j \leqslant x^{\frac{1}{10}}$.

Lemma 9. *Let x be a large even integer; then*

$$P_x(x, x^{\frac{1}{10}}) - \left(\frac{1}{2}\right) \sum_{x^{\frac{1}{10}} < p \leqslant x^{\frac{1}{3}}} P_x(x, p, x^{\frac{1}{10}}) \geqslant \frac{2.6408 x C_x}{(\log x)^2},$$

where $C_x = \prod_{\substack{p|x \\ p>2}} \frac{p-1}{p-2} \prod_{p>2} \left(1 - \frac{1}{(p-1)^2}\right)$.

Proof. In reference [11] take $r(p) = \frac{p}{p-1}$, $K = x$, $Z = x^{\frac{1}{10}}$; then clearly conditions (A_1) and (A_2) of reference [11] are satisfied, and by formula (2.11) of reference [11], we have

$$\Gamma_x(x^{\frac{1}{10}}) = \frac{x}{\varphi(x)} \prod_{p \nmid x} \frac{1 - \frac{1}{p-1}}{1 - \frac{1}{p}} \cdot \frac{e^{-r}}{\log x^{\frac{1}{10}}} \left\{ 1 + O\left(\frac{1}{\log x}\right) \right\}$$

$$= \frac{x}{\varphi(x)} \prod_{\substack{p|x \\ p>2}} \frac{(p-1)^2}{p(p-2)} \prod_{p>2} \left(1 - \frac{1}{(p-1)^2}\right) \frac{e^{-r}}{\log x^{\frac{1}{10}}} \left\{ 1 + O\left(\frac{1}{\log x}\right) \right\} = \frac{20 e^{-r} C_x}{\log x} \left\{ 1 + O\left(\frac{1}{\log x}\right) \right\}. \tag{25}$$

Here r is Euler's constant. When $0 < u \leqslant 2$, let $F(u) = \frac{2e^r}{u}$, $f(u) = 0$. When $u \geqslant 2$, let $(uF(u))' = f(u-1)$, $(uf(u))' = F(u-1)$; when $2 < u \leqslant 3$, $uF(u) = 2F(2)$, $F(u) = \frac{2e^r}{u}$. When $2 < u \leqslant 4$,

$$uf(u) = \int_2^u F(t-1)\, dt = 2e^r \log(u-1), \qquad f(u) = \frac{2e^r \log(u-1)}{u}.$$

When $3 \leqslant u \leqslant 4$, we have

$$uF(u) = 2e^r + \int_3^u f(t-1)\, dt = 2e^r \left(1 + \int_2^{u-1} \frac{\log(t-1)}{t}\, dt\right),$$

and also

$$5f(5) = 2e^r \log 3 + \int_4^5 F(u-1)\,du = 2e^r \left(\log 4 + \int_3^4 \frac{du}{u} \int_2^{u-1} \frac{\log(t-1)}{t}\,dt \right).$$

In Theorem A of reference [11], take $\xi^2 = x^{\frac{1}{2}-\epsilon}$, $q = 1$, $z = x^{\frac{1}{10}}$; then by (25) and formulas (2.19), (4.18), (3.24) of reference [11], we know that for large x,

$$P_x(x, x^{\frac{1}{10}}) \geqslant \frac{2(1-\sqrt{\epsilon})e^{-r}xC_x f(5)}{(\log x)\left(\log x^{\frac{1}{10}}\right)} \geqslant \left\{ \frac{8(1-\sqrt{\epsilon})xC_x}{(\log x)^2} \right\} \left\{ \log 4 + \int_3^4 \frac{du}{u} \int_2^{u-1} \frac{\log(t-1)}{t}\,dt \right\}. \tag{26}$$

Again, in Theorem A of reference [11] take $\xi^2 = \dfrac{x^{\frac{1}{2}-\epsilon}}{p}$, $q = p$, $z = x^{\frac{1}{10}}$; then by (25) and formulas (2.18), (3.24), (4.18) of reference [11], we have

$$\sum_{x^{\frac{1}{10}} < p \leqslant x^{\frac{1}{3}}} P_x(x, p, x^{\frac{1}{10}}) \leqslant \left\{ \frac{20(1+\sqrt{\epsilon})e^{-r}xC_x}{(\log x)^2} \right\} \left\{ \sum_{x^{\frac{1}{10}} < p \leqslant x^{\frac{1}{5}}} \left(\frac{2e^r}{p}\right) \left(1 + \int_2^{4-\frac{10\log p}{\log x}} \frac{\log(t-1)}{t}\,dt\right) \left(\log x^{\frac{1}{10}}\right.\right.$$

$$\left.+ \sum_{x^{\frac{1}{5}} < p \leqslant x^{\frac{1}{3}}} \frac{2e^r \log x^{\frac{1}{10}}}{p \log \dfrac{x^{\frac{1}{2}}}{p}} \right\} \leqslant \left\{ \frac{(4+5\sqrt{\epsilon})xC_x}{\log x} \right\} \left\{ \int_{x^{\frac{1}{10}}}^{x^{\frac{1}{5}}} \frac{dS}{S(\log S)\left(\log \dfrac{x^{\frac{1}{2}}}{S}\right)} \int_2^{4-\frac{10\log S}{\log x}} \frac{\log(t-1)}{t}\,dt + \int_{\frac{1}{10}}^{\frac{1}{3}} \frac{d\alpha}{\alpha \cdots}\right.$$

Let $4 - 10\alpha = u - 1$, $\alpha = \dfrac{5-u}{10}$, so $\dfrac{d\alpha}{\alpha\left(\dfrac{1}{2}-\alpha\right)} = -\dfrac{10\,du}{u(5-u)}$; when $\alpha = \dfrac{1}{10}$, $u = 4$, and when $\alpha = \dfrac{1}{5}$, $u = 3$; hence

$$\int_{\frac{1}{10}}^{\frac{1}{3}} \frac{d\alpha}{\alpha\left(\dfrac{1}{2}-\alpha\right)} \int_2^{4-10\alpha} \frac{\log(t-1)}{t}\,dt = \int_3^4 \frac{10\,du}{u(5-u)} \int_2^{u-1} \frac{\log(t-1)}{t}\,dt.$$

It is clear that, for $1 \leqslant x \leqslant 2$, $\log x \leqslant \dfrac{x-1}{2} + \dfrac{x-1}{1+x}$; hence

$$\int_3^4 \frac{du}{u} \int_2^{u-1} \frac{\log(t-1)}{t}\,dt - \left(\frac{1}{4}\right) \int_{\frac{1}{10}}^{\frac{1}{5}} \frac{d\alpha}{\alpha\left(\dfrac{1}{2}-\alpha\right)} \int_2^{4-10\alpha} \frac{\log(t-1)}{t}\,dt$$

$$= \int_3^4 \left(\frac{1}{u} - \frac{2.5}{u(5-u)}\right) du \int_2^{u-1} \frac{\log(t-1)}{t}\,dt \geqslant \int_3^4 \left\{\frac{2.5-u}{u(5-u)}\right\} du \int_2^{u-1} \left(\frac{t-2}{2} + \frac{t-2}{t}\right)\left(\frac{dt}{t}\right)$$

$$= \int_3^4 \left\{\frac{2.5-u}{2u(5-u)}\right\} \left(u - 3 + \frac{4}{u-1} - 2\right) du = \int_3^4 \left(\frac{1}{2} - \frac{2.25}{u} - \frac{1}{4(5-u)} + \frac{0.75}{u-1}\right) du$$

$$= \frac{1}{2} - 2.25\log\frac{4}{3} - \frac{\log 2}{4} + 0.75\log\frac{3}{2} = \frac{1}{2} + 0.75\log\frac{9}{8} - 1.5\log\frac{4}{3} - \frac{\log 2}{4}$$

$$\geqslant 0.588335 - 0.6048075 = -0.0164725. \tag{16}$$

By (26) and (27), we have

$$P_x(x, x^{\frac{1}{10}}) - \left(\frac{1}{2}\right) \sum_{x^{\frac{1}{10}} < p \leqslant x^{\frac{1}{3}}} P_x(x, p, x^{\frac{1}{10}}) \geqslant \left(\frac{(8 - 50\sqrt{\epsilon})xC_x}{(\log x)^2}\right) \left(\log 4 - \frac{\log 8}{2} - 0.0164725\right) \geqslant \frac{(8xC_x)(0.3}{(\log x)}$$

This proves Lemma 9. □

3 Results

Clearly, we have

$$P_x(1,2) \geqslant P_x(x, x^{\frac{1}{10}}) - \left(\frac{1}{2}\right) \sum_{x^{\frac{1}{10}} < p \leqslant x^{\frac{1}{3}}} P_x(x, p, x^{\frac{1}{10}}) - \frac{\Omega}{2} - x^{0.91}. \tag{17}$$

From (28), Lemma 8, and Lemma 9, we obtain the proof of Theorem 1:

$$(1,2) \text{ holds, and } P_x(1,2) \geqslant \frac{0.67xC_x}{(\log x)^2}.$$

An entirely analogous method gives the proof of Theorem 2.

Acknowledgement. The author wishes to express his heartfelt thanks to Comrades Min Sihe and Wang Yuan for their assistance.

References

[1] Renyi, A., 1948 *Izv. Akad. Nauk SSSR Ser. Mat.*, **2**, 57–78.

[2] Pan, C. D. (Pan Chengdong), 1962 *Acta Math. Sinica*, **12**, 95–106.

[3] Barban, M. B., 1961 *Dokl. Akad. Nauk UzSSR*, **8**, 9–11.

[4] Wang, Y. (Wang Yuan), 1962 *Sci. Sinica*, **11**, 1033–1054.

[5] Pan, C. D. (Pan Chengdong), 1963 *Sci. Sinica*, **12**, 455–473.

[6] Barban, M. B., 1963 *Mat. Sb.*, **61**, 418–425.

[7] Buchstab, A. A., 1965 *Dokl. Akad. Nauk SSSR*, **162**, 739–742.

[8] Vinogradov, A. I., 1965 *Izv. Akad. Nauk SSSR Ser. Mat.*, **29**, 903–934.

[9] Bombieri, E., 1965 *Mathematika*, **12**, 201–225.

[10] Chen, J. R. (Chen Jingrun), 1966 *Kexue Tongbao*, **17**, 385–386.

[11] Richert, H. E., 1969 *Mathematika*, **16**, 1–22.